

Course Name: JAVA PROGRAMMING	
Semester IV	
Course Code	BCA-DC401
Course Title	JAVA PROGRAMMING
Number of Credits	Total: (L: 2 T: 1 P:2)
Total Number of teaching hours	40 Hours
Pre-requisites	Object Oriented Concepts
Course Category	Core

Rationale:

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcome:

- ❖ To make students well versed with programming in java.
- ❖ To understand the use of object oriented features along with their applications.

Detailed Contents

Unit	Title	Hours
Unit 1	Basics of Java programming	12
1.1	Java Programming: History and Features of Java, C++ vs. Java, JDK, JRE, and JVM (Java Virtual Machine), Java Tokens: Data types, Literals and Variables, Operators, Control Structures, Arrays: 1D, 2D and Multidimensional Arrays, Array Class, Concept of Dynamic Arrays Classes: Fundamentals, objects, Creating a class, methods, Access modifiers, Constructors, this keyword, static (variable, method, block), final keyword - final variables, Wrapper Classes, Garbage Collection	6
1.2	Polymorphism: Method overloading, constructor overloading.	6

	Inheritance: Super class, subclass, Types of Inheritance: Single-level Inheritance, Multi-level Inheritance, Hierarchical Inheritance, Multiple Inheritance using interface, Hybrid Inheritance, super keyword, method overriding, use of final methods & classes), abstract class Interfaces: Creating and implementing an interface, difference between abstract class & interface Packages: Concept, Importing a package and creation of the custom package.	
UNIT 2	Exception Handling and Multithreading	8
2.1	Exception Handling: Exception Class, built in checked and unchecked exceptions, user defined exceptions, use of try, catch, throw, throws, finally.	3
2.2	Multi threaded programming: Overview, comparison with multiprocessing, Thread class and runnable interface, life cycle, creation of single and multiple threads, thread priorities, overview of Synchronization.	3
2.3	Java Library: String handling (only main functions): String class, StringBuffer class and StringBuilder class and comparison among them.	2
UNIT 3	Streams, GUI Programming & Event Handling	12
3.1	Streams & Elementary concepts of Input/Output: Byte and Character streams, reading & writing using byte & character streams, System.in and System.out, print and println, reading from a file and writing in a file.	3
3.2	Applets: Introduction and life cycle. AWT controls: Button, Label, TextField, TextArea, Choice lists, list, scrollbars, check boxes, Layout managers Swings: Introduction, Components (JLabel, ImageIcon, JButton, JToggleButton, JCheckBox, JRadioButton, JTextField, JTextArea, JPasswordField, JList, JComboBox, JScrollPane), implementation and comparison with AWT.	6
3.2	Elementary concepts of Event Handling: Delegation Event Model, Event classes, interfaces and listeners, Adapter classes, Inner classes.	3
UNIT 4	Networking, Database and Servlets	8
4.1	Networking Basics: Socket (datagram and TCP/IP based client and server socket), factory methods, InetAddress JDBC: JDBC Architecture, JDBC Drivers, Connection, Statement, PreparedStatement, ResultSet, Connecting to the Database.	5
4.2	Introduction to Java Servlets: Life cycle, Interfaces and classes in javax, servlet package (only description), Creating a simple servlet.	3

List of Practical:

List of practicals will be prepared as per subject teacher.

Learning Approach:

Students will be well versed with programming in java and understood the use of object oriented features along with their applications.

References/suggested learning resources:

Books

TEXT BOOKS:

- [T1] Patrick Naughton and Herbert Schildt, "Java-2 The Complete Reference", TMH.
- [T2] Y. Daniel Liang, "Introduction to Java Programming, Comprehensive Version, 7/e"
Pearson

REFERENCE BOOKS: -

- [R1] Krishnamoorthy R, Prabhu S , "Internet and Java Programming", New Age Intl.
- [R2] David Flanagan, Jim Farley, William Crawford and Kris Magnusson, "Java Enterprise in a Nutshell", O'Reilly.

Course Name: C# Programming	
Semester IV	
Course Code:	BCA-EC402
Course Title:	Elective - I C# Programming
Number of Credits:	Total: (L: 3 T:1 P: 2)
Total Number of teaching hours	40 Hours
Pre-requisites	Basic Programming Language
Course Category:	Elective

Rationale:

Microsoft designed it as the official language of its .NET framework. Anything written in the .NET framework runs in Windows, which makes C# one of the official languages of Windows development. With the introduction of .NET Core, C# can now be used to create apps on macOS, Linux, and even Raspberry Pi. Microsoft is constantly adding new features to the language. Microsoft also provided support for seamless integration with other Microsoft technologies, such as Microsoft SQL server, cloud computing, Azure deployment and many more. Having these features makes the language more versatile, easy to learn and increases its usability.

Learning Outcome:

- Understand the concept of DOT NET Framework.
- Understand the management of Data Structure through DOT NET Framework.
- Understand what is Garbage Collection.
- C# Programming language concepts.
- Object Oriented Programming through C#. Use of Inheritance, Multiple & Multi Level Inheritance & Interface.
- Concepts of Abstract Classes & Abstract Functions, properties, Overloading, Over riding, Exception Handling, Database handling, File handling.
- Formatted Input Output handling through C#.
- Windows Application development & Web application development through C#.

Detailed Contents

Unit	Title	Hours
Unit 1	The CLR and .NET Framework	08
1.1	Understand the motivation behind the .NET platform, Common Language Infrastructure (CLI). Know the role of the Common Type System (CTS), the Common Language Specification (CLS) and the Common Language Runtime (CLR)	4
1.2	Understand the assembly, metadata, namespace, type distinction, Contrast single-file and multi-file assemblies, Know the role of the Common Intermediate Language (CIL), Platform independent .NET (Mono / Portable .NET distributions).	4
Unit 2	C# Language Fundamentals	12
2.1	Language Fundamentals, Reference and value Types, primitive types the Nullable and enum types, Classes and objects, defining classes Creating objects, Using static members, Garbage Collector, Overloading Methods, Various Constructors. Encapsulating data, access modifiers, properties, indexers arrays and readonly fields.	7
2.2	Handling errors and throwing exceptions The Root object class. Inheritance and polymorphism specialization and generalization, Abstract classes, nesting of classes. Structures. String and DateTime classes.	5
Unit 3	Advanced Features	10
3.1	Event handling paradigm: Delegates and events. Anonymous delegates and lambda expression FUNC and Action delegates.	3
3.2	Generics Collections Interfaces, overriding interface implementation. Explicit interface implementation. Collection, IEnumerable, IEnumerator, IList, IComparer and their Generic equivalent. Working with generic List, Stack, Dictionary and Queue.	3
3.3	Programming Window Forms Applications: The notifies - subscribers paradigm for handling events. .NET framework for handling GUI events. Introduction to WPF and building an WPF application.	4
Unit 4	Introducing LINQ and XML	10

4.1	XML A quick introduction. LINQ and C#. Defining and executing a Query. Implicitly typed local variables. Anonymous Types, Extension Methods and Lambda Expressions. Putting LINQ to work. LINQ to SQL Fundamentals of ADO.NET Updating retrieving and deleting data using LINQ to SQL.	10
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List Of Practicals :

1. To understand the Visual Studio 2010, its features, installation and overview.
2. Program to show the use of WriteLine(), Write(), ReadLine() and ReadKey() methods.
3. Program to perform various Arithmetic operations demonstrating use of switch statement.
4. Program to print sum of digits of a number, reverse the number and print multiplication table of the number (Use for and do-while loop).
5. Program to print Prime numbers in a user-defined range.
6. Program to print Fibonacci series of n numbers using recursion.
7. Program to print Factorial of a given number with and without using recursion.
8. Program to convert English Distance (Feets & Inches) to Metric Distance (Meters & Centimeters) and vice-versa using properties.
9. Program to declare a class EnglishDistance to store distance in Feets and Inches. Define necessary Constructors, Properties and Methods to find sum and difference of two distance objects.
10. Program to declare a class Date_Time to store date and time entered by user. Check whether user has entered valid date and time, and display message accordingly. Make use of necessary Constructors and properties.
11. Program to declare a class to store a Person's info such as ID, Name, Phone Number, DOB and Address. Use necessary Methods and Properties for taking values from user and setting up the values respectively. Override ToString() method to print the values to the screen.
12. Program to show the use of Abstract classes. Create Abstract Class Shapes for various shapes and create derived classes for shapes like Circle, Triangle, Rectangle, Square to find Area and Perimeter of the shape. Use Overriding of To String Method for printing the Data of Shape.
13. Program to implement array using Array class and implement BinarySearch() & Sort() methods.
14. Program to implement Method Overriding. Make use of 'virtual' and 'override' keywords.
15. Program to show the use of Abstract classes.
16. Programs on exception handling.
17. Program to create a Custom (User-Defined) Exception which is thrown when an invalid name is entered by the user. Also write code to catch and handle this Exception.

18. Programs demonstrating LINQ.
19. Program to find Odd & Even numbers in a given array using LINQ.
20. Program to create an XML document.
21. Program to create a Calculator using Windows Form.
22. Mini Project in C# (50% Weightage)

Learning Approach:

1. In this course student will become familiar with a with C# language. This course will help to develop real life projects and Windows Application as well as Web applications.
2. Initially, Basic concepts of DOT NET framework may be taken up, followed by C# Language concepts.

References/suggested learning resources:

(a) Books

1. Jesse Liberty and Donald Xie , “Programming C# 5.0”, O’REILLY.
2. J.G.R. Sathiaselvan, N Sasikaladevi, “Programming with C# .net”, PHI, 2009.
3. Paul J. Deitel, Harvey Deitel, “C# 2008 for Programmers”, Pearson, 3rd Ed., 2010.
4. Joseph Albahari and Ben Albahari, “C# 5.0 in NUTSHELL”, O’REILLY.
5. Herbert Schildt “C# 4.0: The Complete Reference”, McGraw-Hill
6. Anders Hejlsberg Mads Torgersen Scott Wiltamuth Peter Golde, “The C# Programming Language Fourth Edition”, Microsoft Corporation Addison Wesley

(b) Open source software and website address:

1. <https://www.csharp411.com/best-c-web-sites/>
2. <https://dotnet.microsoft.com/en-us/apps/aspnet/web-apps>
3. <https://www.mono-project.com/>